29. Bar Chart with Different Colors

Aim

To write a Python program to display a bar chart of programming language popularity using different colors for each bar.

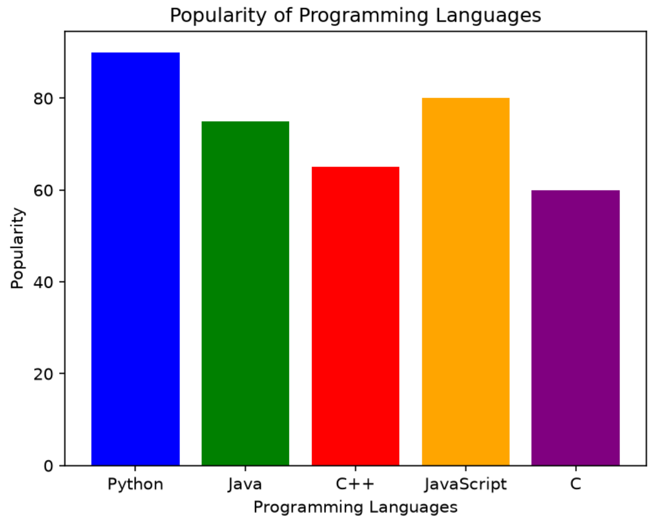
Dataset

Programming Language	Popularity
Python	90
Java	75
C++	65
JavaScript	80
C	60

Python code

```python
import matplotlib.pyplot as plt
languages = ["Python", "Java", "C++", "JavaScript", "C"]
popularity = [90, 75, 65, 80, 60]
colors = ["blue", "green", "red", "orange", "purple"]
plt.bar(languages, popularity, color=colors)
plt.xlabel("Programming Languages")
plt.ylabel("Popularity")
plt.title("Popularity of Programming Languages")
plt.show()
```

Output



Result

A colored bar chart was successfully created to represent programming language popularity.

30. Bar Plot of Scores by Group and Gender

Aim

To write a Python program to create a bar plot showing scores by group and gender using multiple X-values on the same chart.

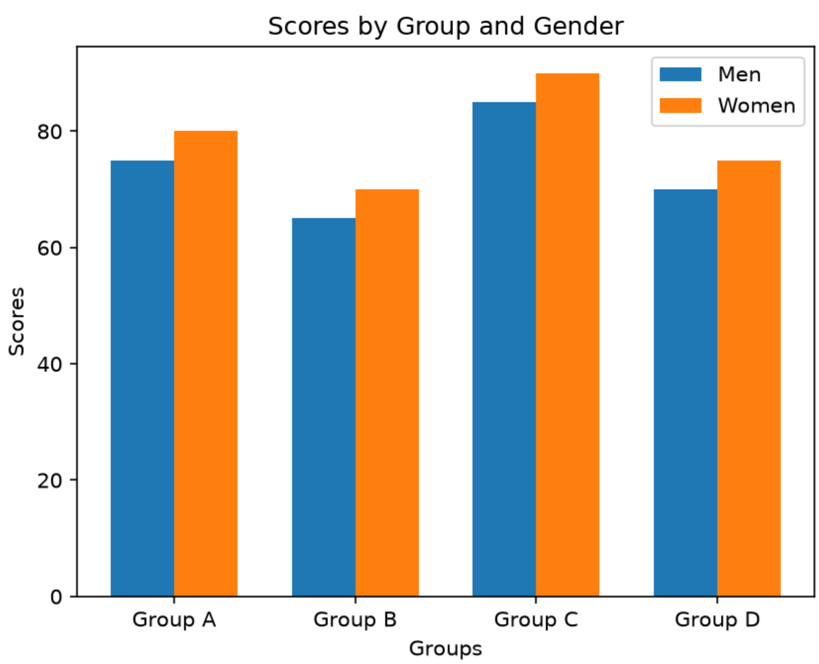
Dataset

Group	Men	Women
Group A	75	80
Group B	65	70
Group C	85	90
Group D	70	75

Python code:

```python
import matplotlib.pyplot as plt
import numpy as np
groups = ["Group A", "Group B", "Group C", "Group D"]
men = [75, 65, 85, 70]
women = [80, 70, 90, 75]
x = np.arange(len(groups))
width = 0.35
plt.bar(x - width/2, men, width, label="Men")
plt.bar(x + width/2, women, width, label="Women")
plt.xlabel("Groups")
plt.ylabel("Scores")
plt.title("Scores by Group and Gender")
plt.xticks(x, groups)
plt.legend()
plt.show()
```

Output:



Result:

The scores of men and women were successfully represented using multiple X-values on the same bar chart.